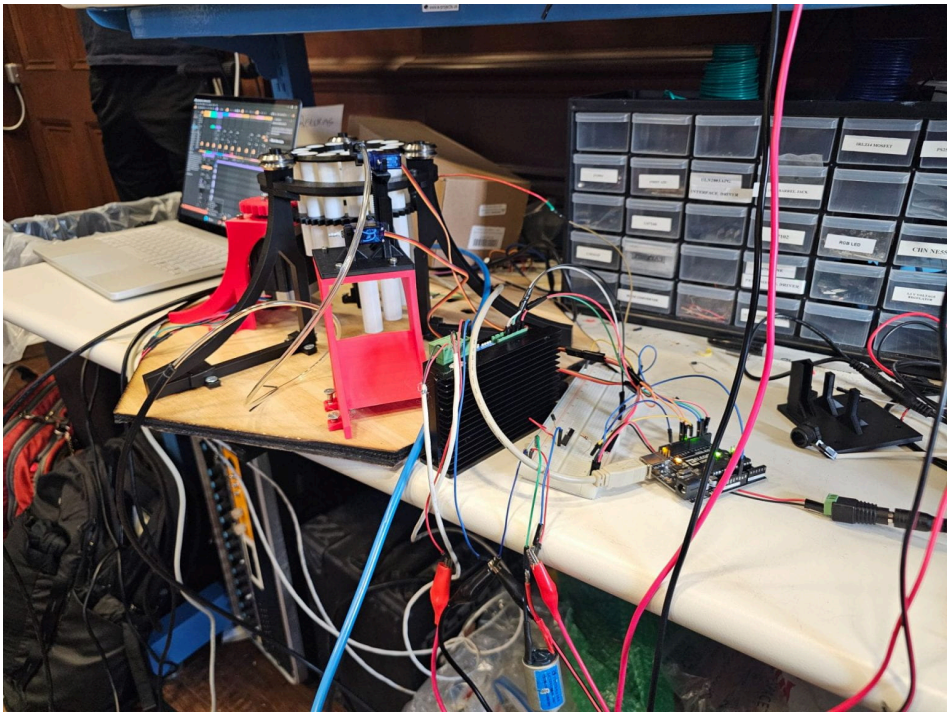
Pan Flute Robot



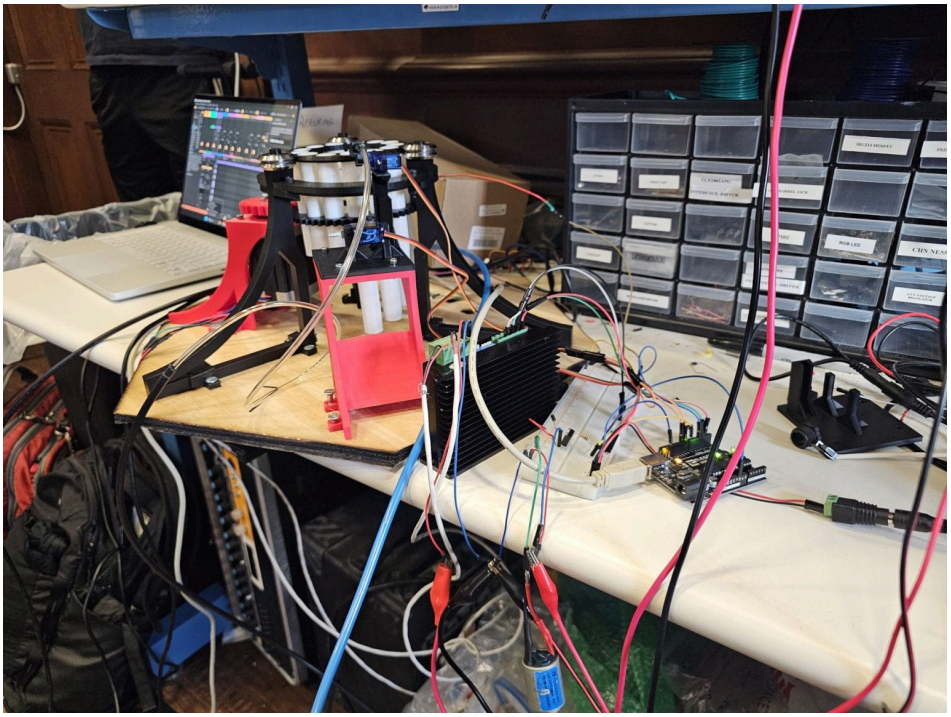
Musical Robotics, B-2025.

William Abraham, Vicente Bastura, Nathan Cote, and Beckett Peale

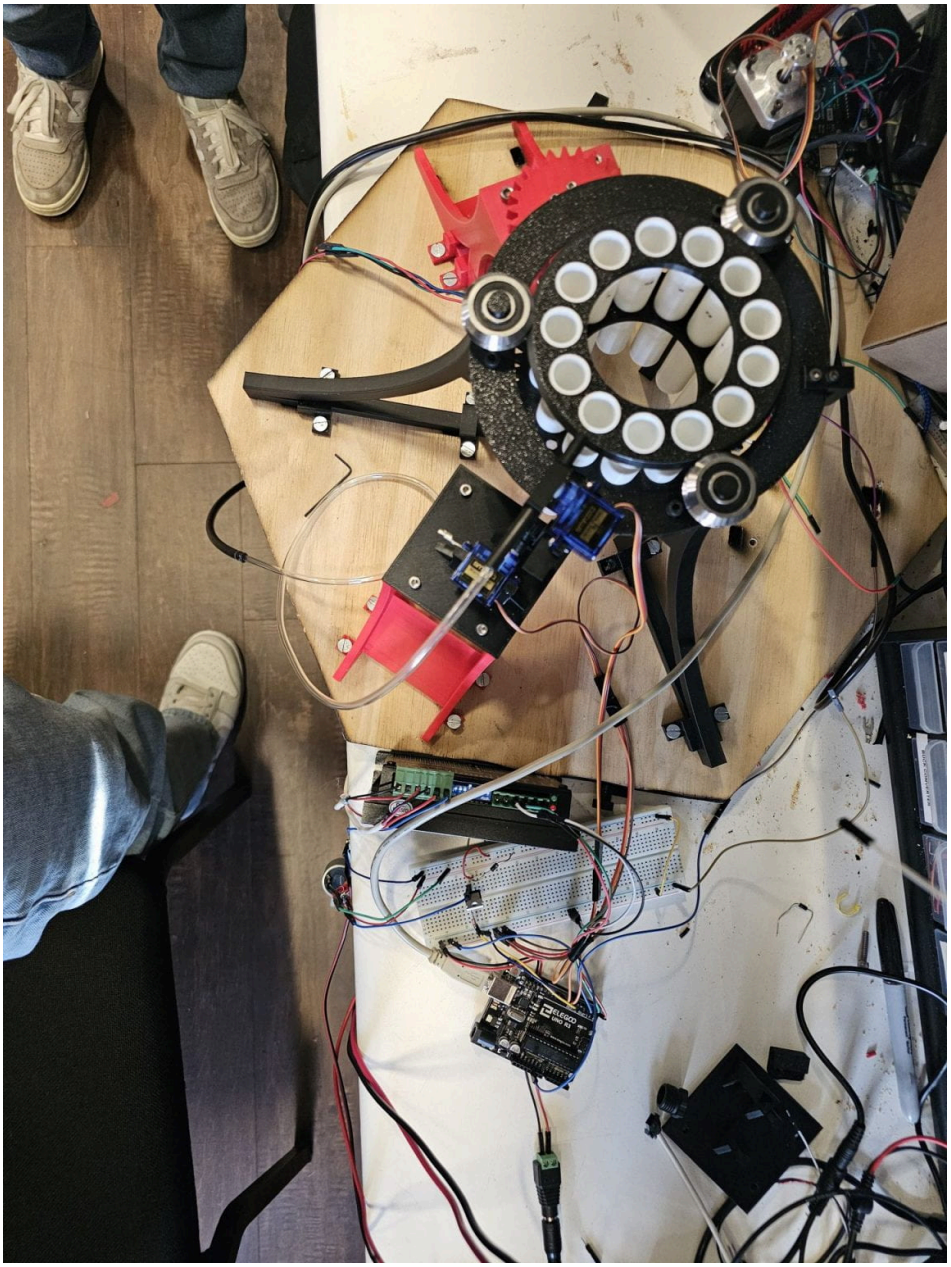
Description

The driving idea for this robot was to take a pan flute and arrange the pipes radially, so they can be rotated and played by one airflow tube. The pan flute “drum” is supported by grooved bearings and connected to a stepper motor via gears. This allows precise turning to each note based on the number of steps between pipes. The blower mechanism blows air into the pipe from a tube, and it has two degrees of freedom from servos and a rack and pinion mechanism. The elevation and angle of the pipe can be adjusted for each individual note using software and allows calibration for desired tone. This personalized calibration grants freedom to the composer for desired tone and feel. A solenoid valve is used to control the airflow on and off. The electrical system has three different powerlines; 5v for servos, 12v for solenoid, and 24v for stepper/driver. All subsystems are connected to an Arduino microcontroller. The programming has two major components. The first being the software to calibrate each notes tone and the second being the main program for playing the instrument. The main program uses serial communication with Ableton Live to receive notes and takes the shortest direction to the desired notes. This allows for easy composition and integration with other instruments.

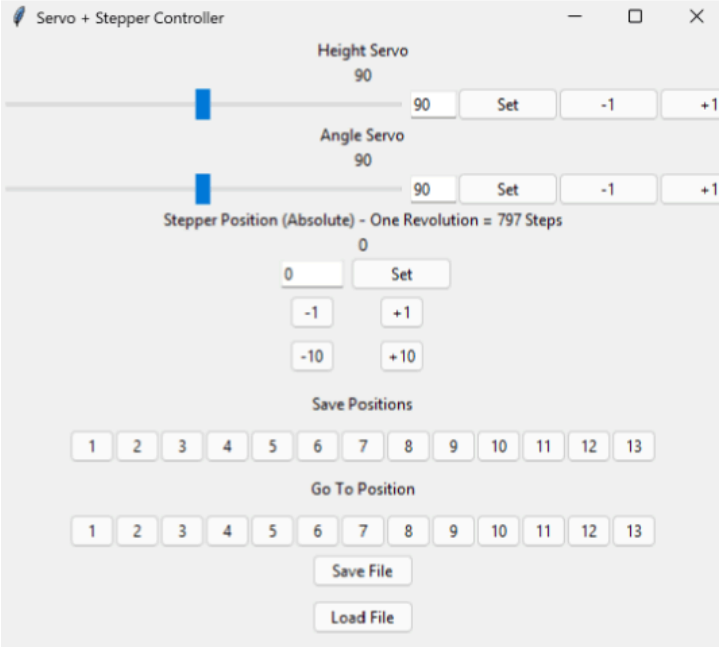
Media



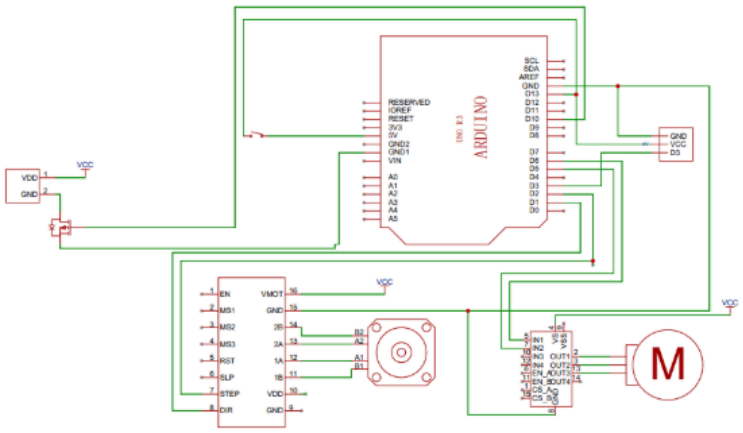
Final Robot



Top View



Calibration Software



Electrical Diagram

Videos